

Figure 6.3 Entropy of the location of the remote after a portion of the living room is searched. If the remote is found (upper right), then no uncertainty remains. Otherwise, entropy is still reduced (lower right), as the distribution of the location of the remote is now narrower.

Either way, the entropy is reduced by looking in the specified portion of the living room. What would happen, then, if you decided to search for the remote in the bathroom? Figure 6.4 shows the corresponding reasoning:

- If the remote is found in the bathroom, then entropy will still drop to zero. However, this is unlikely to happen according to your belief about the location of the remote.
- If the remote is not found in the bathroom, then your posterior belief about the location of the remote doesn't change from Figure 6.2, and the resulting entropy remains the same.

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